

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – FUNCTION CALLING & STRUCTURED OUTPUT TASK

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL2, BL3)	Assessment:	Assessment Tool 3 – Function Calling & Structured Output Task
Weightage:	40% of CIA	Total Marks:	1 * 100 = 100 (1 Question1)
CO2 Statement:	<i>Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems (BL2, BL3)</i>		
Student Name:	Joanathan Packia Singh	Reg. No.:	192472229
Section / Batch:	SLOT A	Date:	03/08/2026

Instructions to Students

- Answer 1 question. Total: 100 Marks.
- Analyze the given scenario and identify the appropriate function(s), tool(s), parameters, and structured output required to solve the task.
- Design a valid function-calling schema with appropriate function names, parameters, data types, required fields, and descriptions based on the given requirements.
- Implement the function-calling workflow demonstrating the complete process: User Query → LLM → Function/Tool Call → Function Execution → Structured Response.
- Ensure that the generated output strictly follows the defined structured format or JSON schema and contains valid, relevant, and consistent information.
- Handle appropriate edge cases such as missing parameters, invalid inputs, ambiguous requests, incorrect data types, or unavailable information, and provide suitable responses without producing invalid function calls.
- Demonstrate the working implementation using suitable test cases, including normal inputs and at least one edge-case scenario.
- Clearly document the designed schema by explaining the purpose of each function, its parameters, data types, required fields, expected input, and expected output.
- Briefly justify the design decisions and explain how function calling and structured output improve the reliability, consistency, and usability of the LLM-based application.

Submission Requirements

Students should submit:

- **Source code / Jupyter Notebook**
- **Function-calling schema**
- **Structured output/JSON schema**
- **Execution screenshots**
- **Test cases and results**
- **Brief documentation explaining the schema and function-calling workflow**

Code of Conduct

I Joanathan Packia Singh (192472229) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Joanathan

26. Sequential Function Calling

A travel assistant receives the request:

“Find a hotel in Chennai for two nights and calculate the total estimated cost.”

Task: Design a workflow in which the LLM calls multiple functions sequentially, such as `search_hotels()` followed by `calculate_cost()`. Show the structured responses at each stage.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Schema Design	30%	Correctly designs a function-calling schema and structured (JSON) output format	Mostly correct schema with minor issues	Schema has significant errors	Schema is fundamentally incorrect or missing
Output Validity	25%	LLM reliably produces valid structured output matching the schema	Mostly valid output with occasional errors	Output is frequently invalid or inconsistent	Output rarely or never matches the schema
Edge-Case Handling	20%	Handles edge cases (missing fields, ambiguous input) gracefully	Handles common cases well	Handles only the simplest cases	Fails on basic cases
Function-Call Flow Demo	15%	Clear demo of the function-call to structured-response flow	Demo covers the main flow	Demo is incomplete	No functional demo presented
Schema Documentation	10%	Well-documented schema design rationale	Adequate documentation	Minimal documentation	No documentation provided

Marks Evaluation:

Question No.	Schema Design(30%)	Output Validity (25%)	Edge-Case Handling (20%)	Function-Call Flow Demo(15%)	Schema Documentation (10%)	Total Marks (100)
Q26						
Overall Total (100)						